



Aim:- Data Visualization II

1. Use the built-in dataset 'titanic' as used in previous problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not.
2. Write observations on the inference from above statistics.

Objective:-

1. To perform exploratory data analysis visualization
2. To understand how age distribution differs among male and female passengers
3. To analyse survival patterns based on age & gender using box plots
4. To identify outliers and variability in age data

Theory

What is Data Visualization?

- Data visualization is the process of converting raw data into visual elements like graphs, plots & charts. It aids in identifying patterns, trends & insights in complex dataset.

- Titanic Dataset

The Titanic dataset is a builtin dataset available in libraries like Seaborn. It contains passenger data from the Titanic.

- age (numerical): passenger's age
- Sex (Categorical): male/female
- Survived (Binary Categorical): 0 = No, 1 = Yes

- Box plot

A box plot (also known as a box-and-whisker plot) visualizes

- Minimum value
- First quartile (Q_1)
- Median (Q_2)
- Third quartile (Q_3)
- Maximum value
- Outliers

It is highly effective in identifying spread, central tendency and outliers in a dataset.

- Outliers

Outliers are data points that deviate significantly from the rest of the data in a dataset. They are values that are usually high or low, compared to the majority of other values.



- Why Do Outliers Matter

- outliers can
- skew your results
- Mislead data analysis & prediction models
- Indicate data entry errors or special case worth further investigation

Algorithm Procedure

1. Import necessary library such as pandas, seaborn, matplotlib
2. Load the Titanic dataset using sns.load_dataset('titanic')
3. Clean the dataset by removing rows with missing value in 'age' or 'sex'
4. Plot a boxplot with
 - x-axis = sex
 - y-axis = age
 - hue = Survived
5. Add appropriate titles & labels
6. Analyse the plot for insights

- Observation



- The bar plot shows different distributions of age for male & female passengers
- female survivors had a higher concentration of younger ages compared to non-survivors
- male passengers showed a wider age range with several visible outliers
- children had a better chance of survival, indicated by smaller age median in the survived category

Applications

- Helps in feature selection for machine learning model
- useful in risk profiling for passenger safety
- can be applied in survival analysis in other domains like healthcare

Conclusion:

In this experiment, we successfully visualised the distribution of age by sex and survived state using a box plot. The visualisation helped me in identifying trend and patterns such as higher survived rate among female younger passengers had higher survival chances etc.

Outcome:

Upon completion of this experiment students are able to understand the usage of box plot in visualising numerical data they gained insight into age-based survival patterns.

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Aim:- Data visualization III.

Download the Iris flower dataset or any other dataset into a Datagame (eg iris) Scan the dataset & give the inference as:-

1. List down the features & their types (eg numeric available in the dataset)
 2. Create a histogram for each dataset feature in the dataset & illustrate the feature distribution
 3. Create a box plot for each feature in the dataset
 4. Compare distributions & identify outliers
- Objective
1. To explore & visualize the structure of the iris dataset
 2. To understand the distribution of numerical features
 3. To identify & interpret outliers
 4. To classify feature types

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Theory

The dataset contains 150 observations of iris flowers from three species

- Iris - setosa

- Iris - versicolour

- Iris - virginica

Each sample has four features

- sepal length (cm) (numeric)

- sepal width (cm) (numeric)

- petal length (cm) (numeric)

- petal width (cm) (numeric)

- species (Categorical / Nominal)

Histogram

A histogram shows the frequency distribution of a single feature. It is useful to visualise shape of the distribution

- skewness

- central tendency

Boxplot

A box plot visualises

- minimum value

- first quartile (q1)

- median (q_2)
- third quartile (q_3)
- maximum value
- outliers

It is highly effective in identifying spread - central tendency & outliers in a dataset

Procedure

1. Download the iris dataset or load it using `seaborn`
2. Import necessary libraries (`pandas`, `seaborn`, `matplotlib`)
3. Load the dataset into a `pandas` dataframe
4. Load basic information using `.head()`, `info()` & `describe()`
5. Create histogram for each of feature
6. Analyse findings & document observations

Observations

1. Feature Types

- All 4 features (sepal length, sepal width, petal length, petal width) are numerical



Histogram Tolerance

- sepal length: Normal distribution
- sepal width: slightly skewed
- petal length & width: well-separated for different species, especially setosa
- petal length is highly effective for classification

Box Plot Tolerance

- sepal width has noticeable outliers
- petal features show good separation for each species
- petal measurements are more consistent than sepal measurements

Outliers

- found mainly in sepal width
- Some outliers could be measurement errors or genus deviation

Applications:

- used in classification models for plant species
- valuable for training machine learning algorithms
- helps in dimensionality reduction & feature selection
- Flower - would example of multivariate visualization



Conclusion:

Through this experiment, we were able to successfully identify & describe all features in the iris data and visualize distributions using histogram & boxplot also able to detect & analyze outliers and compare feature variability across the clusters.

Outcome

Upon completion of this experiment, students are able to understand the usage of boxplot in visualizing numerical data they gained insights into pattern plotting.